

Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

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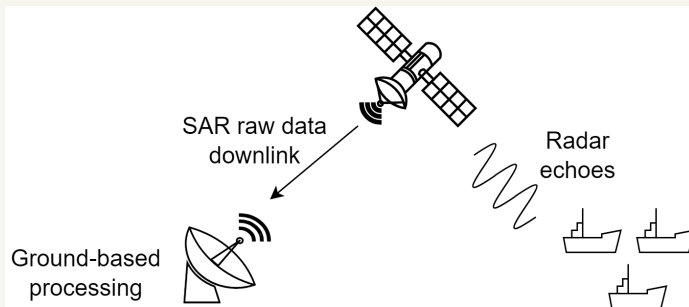
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Outline

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- ▶ System Definition
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Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.

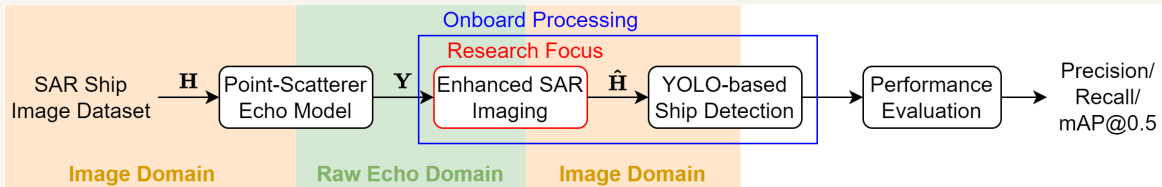


Problem and Motivation

- ▶ Ground-Based Processing
 - ▶ Large raw-data volume
 - ▶ Limited downlink bandwidth
 - ▶ Delayed target information
- ▶ Onboard Processing
 - ▶ Reduced data transmission and detection latency
 - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
 - ▶ Lightweight SAR processing for low-latency ship detection

Evaluation Framework

- ▶ Research Focus: Enhanced SAR imaging
- ▶ Evaluation Task: YOLO-based ship detection



▶ Evaluation Metrics

- ▶ Precision: $\frac{TP}{TP+FP}$
- ▶ Recall: $\frac{TP}{TP+FN}$
- ▶ mAP@0.5: Mean AP across classes at IoU threshold = 0.5

TP: True Positive, FP: False Positive, FN: False Negative

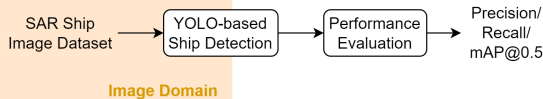
Dataset and Detection Model Setup

► HRSID

- High-Resolution SAR Images Dataset
- Training: 3,642 images
- Validation: 1,962 images
- Image size: 800×800 pixels

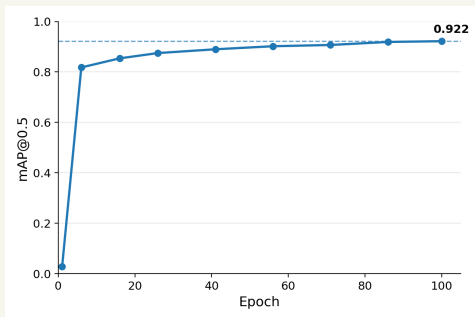
► YOLO Detection Model

- YOLO11m fine-tuned on HRSID (100 epochs)



► Validation Performance

- Precision: 0.910
- Recall: 0.840
- mAP@0.5: 0.922

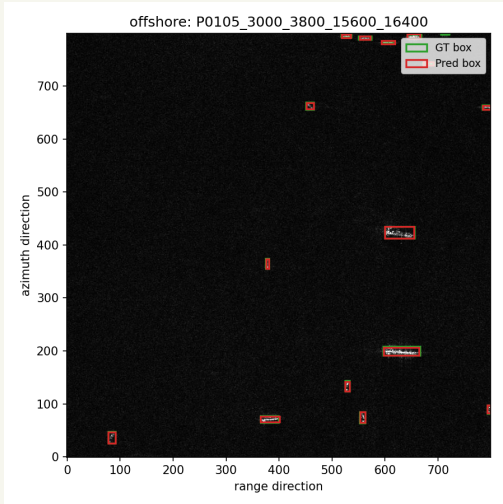


YOLO Ship Detection Results (1)

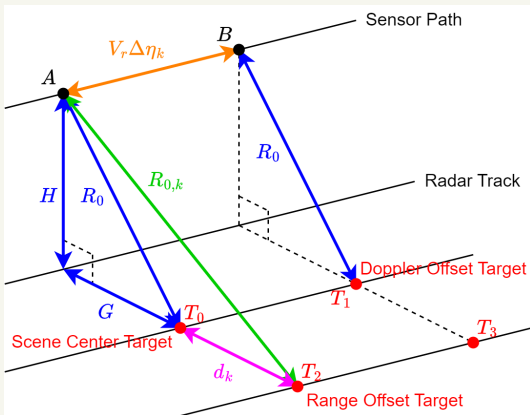
	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

- Offshore scenes achieve better detection performance than inshore scenes.

YOLO Ship Detection Results (2)



Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $$R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta\eta_k)^2 V_r^2}$$
 - $R_{0,k} \approx R_0 + d_k$
 - R_0 : slant range to scene center
 - d_k : ground-range offset of scatterer k , approximated as slant-range offset
 - $\eta_m = m\Delta\eta$: slow time
 - $\Delta\eta_k$: azimuth-time offset of scatterer k

Point-Scatterer Echo Model: Signal Synthesis

► Geometry-Dependent Delay and Phase

$$\text{► } \tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}, \phi_k(\eta_m) = -4\pi \frac{R_k(\eta_m)}{\lambda}$$

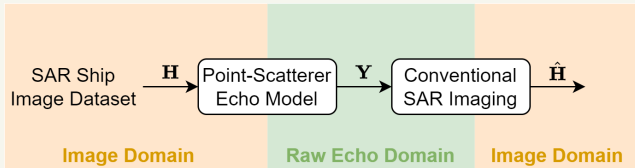
► Echo Synthesis: $\mathbf{S} = [s_{m,n}]_{M \times N}$

$$s_{m,n} = \sum_k A_k \operatorname{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r (\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

► $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time

► A_k : scattering coefficient of scatterer k , $K_r = \frac{B_r}{T_r}$

Conventional SAR Imaging



- Forward model: $\mathbf{Y} = \mathcal{E}(\mathbf{H}) + \mathbf{N}$
 - $\mathcal{E}(\cdot)$: echo formation operator, $\mathbf{N}$: additive thermal noise
- Imaging goal: Recover $\mathbf{H}$ from $\mathbf{Y}$.
- Conventional methods: RDA, CSA, and BP

Regularization for SAR Imaging

► Forward model: $\mathbf{Y} = \mathcal{E}(\mathbf{H}) + \mathbf{N}$

► Regularized reconstruction:

$$\hat{\mathbf{H}} = \arg \min_{\mathbf{H}} \underbrace{\frac{1}{2} \|\mathbf{Y} - \mathcal{E}(\mathbf{H})\|_F^2}_{\text{Data fidelity}} + \lambda \underbrace{\mathcal{R}(\mathbf{H})}_{\text{Regularization}}$$

► $\mathcal{R}(\mathbf{H})$: prior knowledge of the SAR image

► Sparse prior $\rightarrow$ Compressive Sensing (CS)-based SAR imaging

Sparse Regularization

Regularization	$\mathcal{R}(\mathbf{H})$	Property
ℓ_0	$\ \mathbf{H}\ _0$	Exact sparsity; non-convex
ℓ_1	$\sum_{i,j} H_{i,j} $	Convex; tractable
$\ell_{1/2}$	$\sum_{i,j} H_{i,j} ^{1/2}$	Stronger sparsity; non-convex
TV	$\text{TV}(\mathbf{H})$	Spatial smoothness / edge preservation

- Current implementation: ℓ_1 -regularized reconstruction

Prediction Result

Slicing

Conclusion

Reference